Nico Gerardo Bers

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Winchester, MA & Evanston, IL | Last updated October 14, 2026

Education

- **Northwestern University** *Sept. 2023 - June 2027*
  B.A. in Physics with a concentration in Astronomy; Minor in Machine Learning and Data Science Evanston, IL
  - GPA: 3.90/4.00, *Phi Beta Kappa* (Junior year), *Goldwater Scholar '26*
  - Relevant Coursework: Astrophysical Radiative Processes and Transport (graduate level); Quantum Mechanics; Advanced Electricity and Magnetism 1; Observational Astrophysics; Machine Learning; Classical Hamiltonian and Lagrangian Mechanics; Statistical Mechanics; Linear Algebra, Fourier Analysis, and Differential Equations.

Research Experience and Employment

- **Northwestern University Dept. of Physics and Astronomy** *Jan. 2026 - Present*
  Undergraduate Research Assistant in Lena Murchikova's group Evanston, IL
  - Working with Mark Gorski and Lena Murchikova to use machine learning to search for potential accreting multiphased gas structures in the Galactic Center with ALMA radio telescope data.
- **University of Melbourne** *July 2025 - Dec. 2025*
  Science Research Project in Andrew Melatos's group, Member of the LIGO Scientific Collaboration Melbourne, Australia
  - Performing a continuous wave search for the Crab Pulsar in real gravitational-wave data from LIGO's third observing run.
  - Tuning adaptive noise cancellation to improve tracking a neutron star's gravitational-wave frequency with a hidden Markov model's Viterbi decoding algorithm in the presence of non-Gaussian interference.
- **Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA) [🌐]** *Sept. 2023 - July 2025*
  Undergraduate Research Assistant in Vicky Kalogera's group, Member of the LIGO Scientific Collaboration Evanston, IL
  - Gravitational-wave research constraining the redshift distribution of the farthest binary black hole mergers, simulating over 1000 black hole collisions with current LIGO detector sensitivity with hierarchical Bayesian inference and the stochastic background.
  - Developed extended classes of BILBY and GWPopulation libraries for new implementations of the Bayesian templated background search for the stochastic gravitational wave background.
- **Veson Nautical [🌐]** *June 2024 - Sept. 2024*
  Data Science Intern Boston, MA
  - Developed a hierarchical machine learning regression model to estimate the fuel consumption of cargo ships in port stoppages. Improved previous model performance by more than 2 times accuracy.
  - Created a voyage routing estimator for optimizing any maritime voyage using Dijkstra and A* algorithms on weighted graphs with hundreds of thousands of edges and nodes.
- **Veson Nautical [🌐]** *June 2021 - Aug. 2021, June 2022 - Aug. 2022*
  Software Development Intern Boston, MA
  - Developed an iOS app in Swift for nautical fleet tracking and accounting software platform.
  - Responsible for integrating Swift with backend via REST APIs, front-end, and data querying and manipulation with GraphQL and SQL.
- **NASA, TSGC, UT Austin [🌐]** *June 2021 - Aug. 2021*
  STEM Enhancement in Earth Science Research Internship, Earth Systems Group Remote
  - Investigated correlations between mosquito populations and land cover types and contributed to satellite imagery classification.

Papers

- **In preparation** *Oct. 2026*
  - **N. Bers**, M. Gorski, & L. Murchikova. *Exploring the Galactic Center with PCA Tomography: Multiphased Gas in the Inner Parsec*. In prep. (2026).
- **Submitted to *Physical Review D* *Corresponding Author*** *Oct. 6, 2026*
  - **N.G. Bers**, T. Kimpson, Y. Bu, N.K.Y. Low, R. Evans, A. Melatos. *Enhancing searches for gravitational waves from the Crab pulsar in LIGO's O3b with adaptive noise cancellation*. Submitted to Phys. Rev. D (2026). LIGO DCC No. LIGO-P2600498.
- **Published in the *Astrophysical Journal*** *Jan 16, 2026*
  - **N. Bers** & A.S. Biscoveanu. *Probing the peak of star formation with the stochastic background of binary black hole mergers*. The Astrophysical Journal (2026) 997: 108. DOI 10.3847/1538-4357/ae2319.

Honors, Scholarships, and Awards

- **NASA Illinois Space Grant Consortium Summer Program** *June - Sept. 2026*
  - Awarded grant for summer program to continue full-time research in Lena Murchikova's group.
- **Northwestern Summer Undergraduate Research Grant** *June - Sept. 2026*
  - Wrote scientific research proposal and awarded grant to further fund summer research in Lena Murchikova's group.
- **Elected *Phi Beta Kappa* 2026** *May 2026*
  Phi Beta Kappa Society, Alpha Chapter of Illinois [🌐]
  - Elected as a Junior, with no more than 2% of the class elected.
- **2026 *Barry Goldwater Scholar*** *Mar. 2026*
  Barry Goldwater Scholarship and Excellence in Education Foundation [🌐]
  - Awarded to sophomores/juniors with exceptional research potential in STEM fields.
- **2025 USRA Distinguished Undergraduate Honorable Mention** *Oct. 2025*
  Universities Space Research Association [🌐]
  - One of five honorable mentions recognizing contributions to science and US space research, chosen from 130 applicants (5 awardees and 5 honorable mentions).
- **2024-2025 Scholar in Physics & Astronomy** *June 5, 2025*
  Northwestern University Department of Physics and Astronomy [🌐]
  - Received a departmental scholar award for academic and research achievements.
- **NASA Illinois Space Grant Consortium Undergraduate Grant Scholarship Recipient** *Sept. 2024 - June 2025*
  NASA Illinois Space Grant Consortium [🌐]
  - Received a NASA ILSGC scholarship for my academic and research pursuits and successes in STEM.
- **Honorable Mention Rapid Fire Research** *May 31, 2024 & June 6, 2025*
  Northwestern University Department of Physics and Astronomy [🌐][🌐]
  - Received two honorable mentions for my presentation of my research in four minutes or less, competing in a field of graduate and undergraduate researchers.

Leadership, Mentoring, and Teaching Experience

- **Student Org. SAGE Co-President (2026-27), Executive Board (2024-26)** *Sept. 2024 - Present*
  Northwestern Survivor Advocacy through Greek Engagement (SAGE) [🌐] Evanston, IL
  - Designed new standards aimed at preventing sexual violence within Greek Life working with administration, raised more than $6,000 for a local rape crisis center, bring speakers to campus, and provide fraternity trainings.
- **Teaching Assistant - Physics 131-A: Mechanics** *Sept. 2026 - Present*
  NU School of Professional Studies / Dept. of Physics and Astronomy Evanston, IL
  - Leading 60 total students in two weekly lab sections and office hours for introductory mechanics.
- **Teaching Assistant - Physics 131-B: Electricity and Magnetism** *Jan. 2026 - Mar. 2026*
  NU School of Professional Studies / Dept. of Physics and Astronomy Evanston, IL
  - Leading two weekly lab sections and office hours for introductory electricity and magnetism with 40 students total.
  - Explaining physical concepts and lab procedure, and grading weekly lab reports and worksheets.
- **Certified Sexual Health and Assault Peer Educator** *Jan. 2026 - Present*
  Northwestern Sexual Health and Assault Peer Educators (SHAPE) Group [🌐]Evanston, IL
  - Completed training with NU Center for Student Advocacy and Wellness for one quarter.
  - Deliver presentations and trainings to other student groups regarding sexual health and violence prevention.
- **CS 214 Data Structures Peer Mentor** *April 2025 - June 2025*
  Northwestern Computer Science Dept. [🌐]Evanston, IL
  - Four weekly office hours; responsible for answering lecture questions and grading.

ORAL AND POSTER PRESENTATIONS O=ORAL, P=POSTER

- **Informal invited presentation at Prof. Erin Kara's MIT Black Hole Group [O]** *Aug. 25, 2026*
  Stellar Mass Black Hole Collisions and the SMBH in Our Galaxy Cambridge, MA
- **Contributed talk at Harvard CfA Astro AI 2026 workshop [O]** *Jun. 18, 2026*
  Ingredients of Our Black Hole's Diet: Unsupervised Learning to Explore Gas Phases in the Milky Way Center Cambridge, MA
- **NU Undergraduate Research Expo [P]** *May 21, 2026*
  Ingredients of Our Black Hole's Diet: Correlating Different Gas Phases in the Center of the Milky Way Evanston, IL
- **NU Computation and Data Exchange (CoDEx) 2026 [P]** *Apr. 21, 2026*
  Exploring the Farthest Black Hole Collisions Using the Gravitational-Wave Background Evanston, IL
- **Invited presentation at Monash University Gravity Group [O]** *Oct. 23, 2025*
  Searching for Continuous Gravitational Waves with Adaptive Noise Cancellation: Featuring the Crab Pulsar Melbourne, AUS
- **Final presentation for Science Research Project course[O]** *Oct. 20, 2025*
  Searching for Continuous Gravitational Waves with Adaptive Noise Cancellation: Featuring the Crab Pulsar Melbourne, AUS
- **American Astronomical Society 246th Meeting [O, non-presenting author]** *June 30, 2025*
  ◦ Biscoveanu, A., & **Bers, N.** (2025). **Probing the peak of star formation with the stochastic background of binary black hole mergers**. Bulletin of the AAS, 57(4).
- **Rapid Fire Research Physics & Astronomy Dept. Northwestern [O]** *June 6, 2025*
  Exploring the farthest black hole collisions using the gravitational-wave background Evanston, IL
- **34th Midwest Relativity Meeting U. Michigan [O]** *Nov. 17, 2024*
  Probing the peak of star formation with the stochastic background of binary black hole mergers Ann Arbor, MI
- **Rapid Fire Research Physics & Astronomy Dept. Northwestern [O]** *May 31, 2024*
  A Symphony of Binary Black Hole Collisions: Studying Cosmic Noon with Gravitational Wave Backgrounds Evanston, IL
- **Northwestern 2024 Undergraduate Research & Arts Expo. [P]** *May 23, 2024*
  A Symphony of Binary Black Hole Collisions: Studying Cosmic Noon with Gravitational Wave Backgrounds Evanston, IL
- **Chicagoland LIGO-Virgo-KAGRA Meeting [O]** *May 14, 2024*
  Probing the Peak of Star Formation with the Stochastic Background of Binary Black Hole Mergers Evanston, IL
- **American Geophysical Union Fall Meeting, Bright Stars [P]** *Dec. 2021*
  An Analysis of the Effect of Land Cover Type on Mosquito Populations Remote
  ◦ Kamara A., **Bers N.**, et al. (2021, December). **An Analysis of the Effect of Land Cover Type on Mosquito Populations**. In *AGU Fall Meeting Abstracts* (VOL. 2021, PP. ED45B-0637).

SKILLS

- **Languages:** Bilingual in English and Spanish, intermediate working Hebrew, limited working French.
- **Programming Languages:** Python, CARTA, Java, SQL, MATLAB, Mathematica, Swift, DSSL2, Racket/Scheme, Git; some experience with HTML, JavaScript, CSS, React Native.
- **Packages and Libraries:** AstroPy, SpectralCube, NumPy, Pandas, Matplotlib, SciPy, BILBY, GWPopulation, NetworkX, Scikit-learn, TensorFlow, XGBoost, CatBoost, LightGBM, Geopandas, Streamlit, and Pydeck. HTCondor, Slurm Workload Manager, High-Performance Computing (HPC) (LIGO Data Grid, NU QUEST).
- **Machine Learning:** Regression and classification, deep neural networks, convolutional neural networks, random forests, principal component analysis, gaussian mixture models, and gradient boosting models.
- **Interests:** Running (5th place in age group for ChiTown half-marathon 2026), astrophotography, rowing (competed in Head of the Charles Regatta in double), tennis, reading, skiing, drawing.